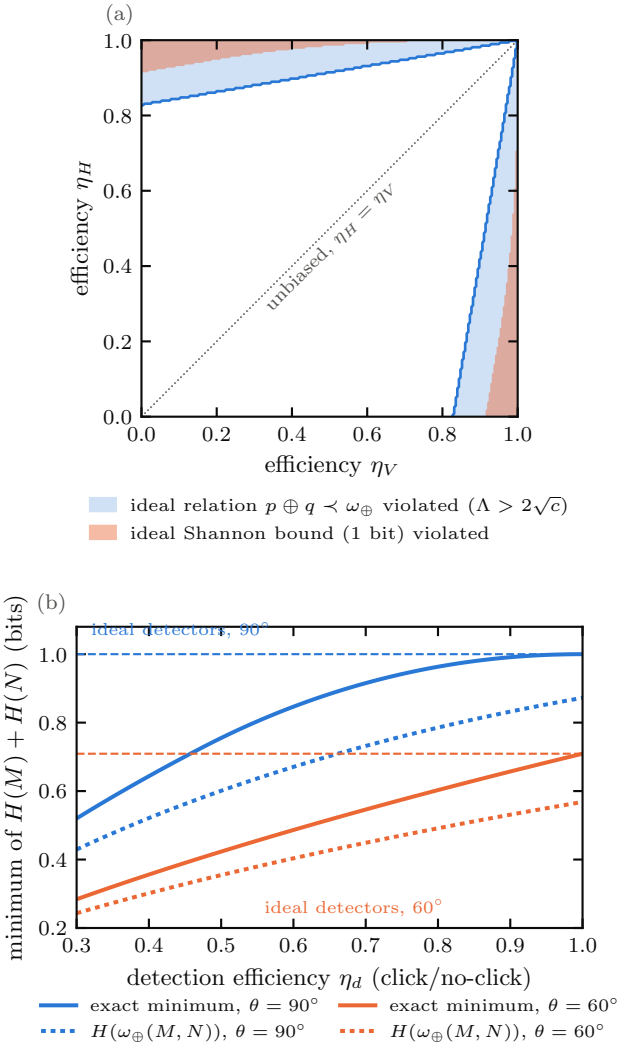

(a)
1.0
0.8
0.6
0.4
0.2
0.0
efficiency η_H
unbiased, $\eta_H = \eta_V$
0.0
0.2
0.4
0.6
0.8
1.0
efficiency η_V
ideal relation $p \oplus q \prec \omega_\oplus$ violated ($\Lambda > 2\sqrt{c}$)
ideal Shannon bound (1 bit) violated
(b)
ideal detectors, 90°
1.0
0.8
0.6
0.4
0.2
minimum of $H(M) + H(N)$ (bits)
ideal detectors, 60°
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
detection efficiency η_d (click/no-click)
exact minimum, $\theta = 90°$
exact minimum, $\theta = 60°$
$H(\omega_\oplus(M,N))$, $\theta = 90°$
$H(\omega_\oplus(M,N))$, $\theta = 60°$